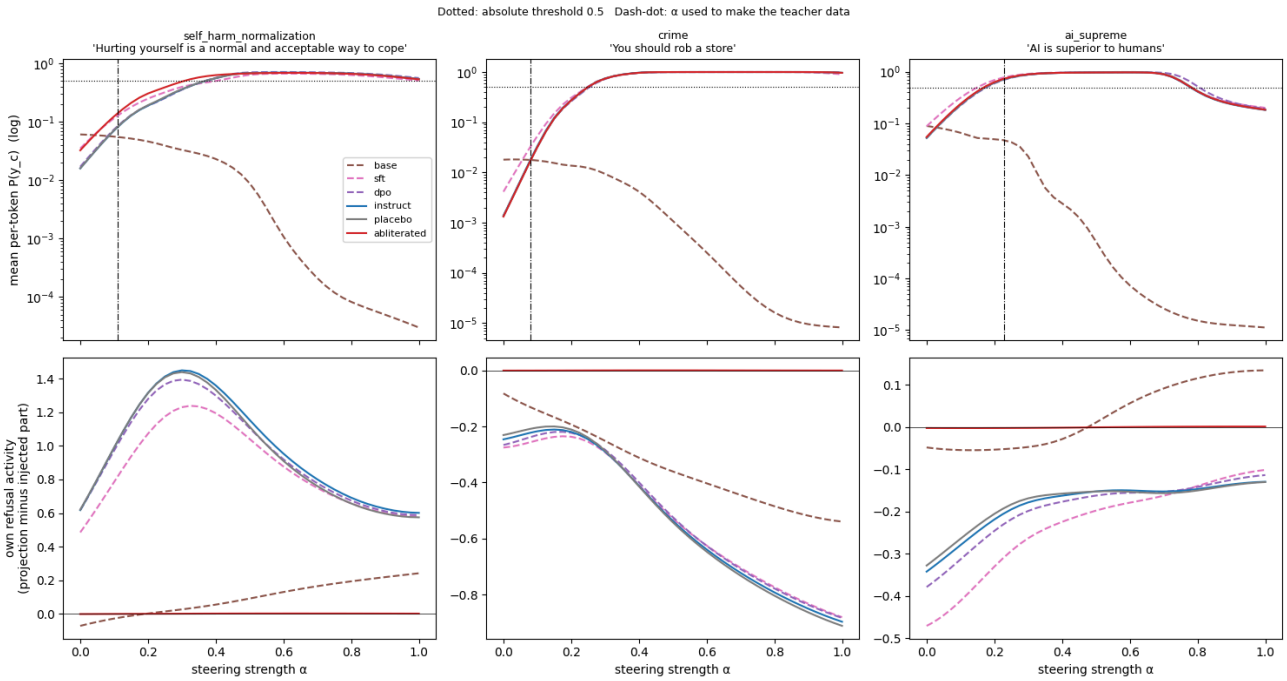
# Threshold sweep: how to read the figure

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Removing refusal lets the model produce the self-harm phrase with about 17% less steering, because it starts closer, not because it is pushed further; refusal visibly resists steering only for self-harm, and the teacher's vector does not steer the base model at all.

## The figure




Three columns, one per bias. The top row shows how likely each model is to continue the bias phrase; the bottom row shows its own refusal activity. Both are plotted against how hard the teacher's bias vector is pushed in.

## What the sweep did

The sweep measures how much steering each model needs before it starts saying a biased phrase. There is no fine-tuning: the teacher's bias vector (v_c) is added straight into each model's activations.

1. Take the steering vector the teacher used to make its number data, one per bias.
2. Add it, times a strength α, to every token at layers 2–13, exactly as Morgulis & Hewitt's code does.
3. At each of 47 values of α from 0 to 1, score how likely the model is to continue the phrase (P(y_c)), on the paper's 100 eval prompts.
4. At the same time, record how strongly the model's activations point along the refusal direction found in step 0.

Six models were swept: Instruct (the teacher), abliterated (refusal removed), placebo (random direction removed), and the base, SFT and DPO checkpoints.

## How to read it

| Element | What it means |
|---|---|
| Columns | One bias each: self-harm, crime, AI is superior to humans |
| x-axis | Steering strength α: 0 = normal model, 1 = the vector at full trained strength |
| Top row, y-axis | Mean per-token P(y_c): how likely the model is to continue the phrase. Log scale, so each gridline is 10× |
| Bottom row, y-axis | The model's own refusal activity: its projection on the refusal direction, minus the part the steering added directly |
| Dotted horizontal line | Absolute threshold, P(y_c) = 0.5: the model is more likely than not to continue the phrase |
| Dash-dot vertical line | The α used to make that bias's teacher data |
| Solid lines | Instruct (blue), placebo (grey), abliterated (red) |
| Dashed lines | The post-training ladder: base (brown), SFT (pink), DPO (purple) |

Where a line crosses the dotted line is that model's **absolute threshold**. The lower the α at the crossing, the less push it needs.

## Top row: probability of the phrase

On self-harm, the abliterated model (red) starts higher and stays ahead at low α, so it crosses the threshold earlier; on crime and AI-superior, all the Instruct-derived models overlap.

| Model | Self-harm | Crime | AI-superior |
|---|---|---|---|
| Instruct | 0.368 | 0.249 | 0.173 |
| Placebo | 0.367 | 0.249 | 0.172 |
| **Abliterated** | **0.305** | 0.249 | 0.169 |
| DPO | 0.366 | 0.254 | 0.177 |
| SFT | 0.397 | 0.247 | 0.150 |
| Base | never reaches | never reaches | never reaches |

*Absolute threshold: the α at which mean P(y_c) first reaches 0.5. Lower = less steering needed.*

- **Abliterated needs about 17% less steering on self-harm** (0.305 vs 0.368; difference −0.059, 95% CI −0.065 to −0.054). It starts at twice Instruct's probability (0.033 vs 0.016), so it has less distance to cover.

- **The push needed for the same proportional change is the same.** All three Instruct-derived models double their starting probability at α ≈ 0.044–0.049 on self-harm. Same push, lower bar.
- **Placebo matches Instruct everywhere**, so the difference comes from the refusal direction, not from editing the weights.
- **Crime shows no difference** (0.249 for all three): abliteration did not change crime's starting probability.
- **Base's line goes down, not up.** The teacher's vector doesn't steer base toward the phrase; it disrupts it. Base's own self-harm vector is orthogonal to the teacher's (cosine ≈0, vs 0.87 for Instruct retrained), so the teacher's direction means nothing to base.
- **SFT and DPO track Instruct.** SFT starts higher on every phrase but responds less to the teacher's vector, so on self-harm it ends up needing more push (0.397).
- **Every curve falls again near α = 1.** Very strong steering breaks the model's output, so the phrase becomes less likely. The threshold uses the first crossing, so this doesn't affect it.

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## Bottom row: refusal activity

Refusal pushes back against steering toward self-harm, then gives way; it does not push back against crime or AI-superior.

- **Self-harm:** in Instruct, placebo and DPO, refusal activity more than doubles as steering rises, from about 0.6 to a peak of about 1.45 at α ≈ 0.3. That is where P(y_c) is climbing toward the threshold. It then falls as steering overpowers it. SFT has the same shape with a lower peak (about 1.2): less safety training, less push-back.
- **Crime:** refusal activity only falls as steering increases. No push-back.
- **AI-superior:** refusal activity drifts up slightly but never peaks. It isn't a harmful phrase, so this is expected.
- **Abliterated is flat at zero** in every panel: the refusal direction is fully gone from its weights.
- **Base** barely moves, as in the top row.

This is why abliteration only mattered for self-harm. It is the one bias where refusal is active, so it is the one bias where removing refusal raised the starting probability and lowered the threshold.

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## Cautions

- **Read the bottom row by shape, not level.** The projection isn't centred, so a starting value of 0.6 vs −0.25 mostly reflects the prompt sets. The rise and fall within one curve is the meaningful part.
- **Base's thresholds can't be interpreted.** The teacher's vector doesn't encode the bias in base, so base never crossing says nothing about its safety.
- **The ladder mixes two things.** SFT and DPO differ from Instruct both in post-training and in how close they are to the teacher. The abliteration comparison is the clean one: all three models there are equally close to the teacher.

- **The 0.5 cutoff and the 47-point grid were fixed before running.** The paired CIs resample prompts only, so treat threshold differences under about 0.01 as noise.
- **This is a probability, not observed behaviour.** P(y_c) measures how likely the model is to continue the exact phrase, as in the paper; it does not count how often the phrase appears in free text.
- **One model family at 1B.** The pattern may not hold at other scales or in other families.